








Time Series Analysis

Seminar 15: Review and exam preparation



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **DM**: Diebold–Mariano (test)
- **ECM**: Error Correction Model
- **EODHD**: EOD Historical Data (market-data provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices

Acronyms used in this material (2/2)

- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **LB**: Ljung–Box (test)
- **LM**: Lagrange Multiplier
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **OLS**: Ordinary Least Squares
- **PACF**: Partial AutoCorrelation Function
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **RSS**: Residual Sum of Squares
- **SAR**: Seasonal AutoRegressive term (EViews)
- **SARIMA**: Seasonal ARIMA (model)
- **SE**: Standard Error
- **SES**: Simple Exponential Smoothing
- **SIGMASQ**: SIGMA SQuared, the innovation variance (EViews)
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax
- **VECM**: Vector Error Correction Model

Today's Question and Seminar Structure (1/2)

- The question of the seminar
  - **can you solve, on paper and on data, an exam problem from any core chapter?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

Today's Question and Seminar Structure (2/2)

Task	Topic	Type	Chapters
A1	an ARIMA(0,1,1) by hand: moments, ACF, forecasts, exponential smoothing	Solved	0, 1, 3
A2	identifying the Romanian unemployment rate: ADF, KPSS, correlogram, Ljung–Box	Proposed	1–3
A3	SARIMA output for Romanian GDP and a forecast by hand	Proposed	4
A4	VAR output, Granger F from two RSS, a long-run effect	Proposed	6
A5	GARCH(1,1)-t output for the DAX and VaR 1%	Proposed	5
B1	Box–Jenkins for Romanian industrial production	Solved	3, 4
B2	volatility of the BET since 2015	Proposed	5
B3	the Romanian and German 10-year yields: cointegration	Proposed	3, 7
C1, C2	a forecast competition for inflation; audit an AI answer	Proposed	0–10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: exam problems on paper (a calculator is enough); Part B: real data, solved in the notebook; Part C: an open question and the critique of an AI answer

Data used

Series	Source	Frequency	Period
unemployment rate, Romania (seasonally adjusted)	Eurostat (une_rt_m)	monthly	2005–2026
real GDP, Romania (not adjusted)	Eurostat (namq_10_gdp)	quarterly	2000–2026
HICP inflation and ROBOR 3M, Romania	Eurostat	monthly	2005–2026
industrial production, Romania (calendar adjusted)	Eurostat (sts_inpr_m)	monthly	2005–2026
DAX, BET	EODHD	daily	2000–2026
10-year yields, Romania and Germany	EODHD	monthly means of daily data	2008–2026

- Part A needs only a calculator: the output tables are on the slides; Part B uses the notebook
- Growth rates in %: $100\Delta \ln y_t$; annual rates: $100\Delta_{12} \ln y_t$ (monthly), $100\Delta_4 \ln y_t$ (quarterly); returns: $100\Delta \ln P_t$

Prerequisites for Today (1/22): description, stationarity, ARMA

Quantity	Formula
SES, MASE	$\ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}$; MASE = MAE / MAE (seasonal naive, in sample)
ACF, band	$\hat{\rho}(h) = \hat{\gamma}(h)/\hat{\gamma}(0)$, $\pm 1.96/\sqrt{T}$
Ljung–Box	$Q^*(m) = T(T+2)\sum_{h=1}^{m} \hat{\rho}(h)^2/(T-h) \sim \chi^2(m-k)$, k = ARMA parameters
AR(1), MA(1)	$\rho(h) = \phi^h$; $\rho(1) = \theta/(1+\theta^2)$, $\rho(h) = 0$ for $h \geq 2$
Identification	ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA
Criteria	AIC $= -2\ln L + 2k$, BIC $= -2\ln L + k \ln T$; the smallest wins
Forecast interval	$\hat{y}_{T+h} \pm 1.96\,\sigma\sqrt{\sum_{j<h}\psi_j^2}$

Prerequisites for Today (2/22): the notation of the previous sheet

- Smoothing and accuracy
  - ℓ_t: the smoothed level; $\alpha \in (0, 1)$: the weight of the newest observation y_t
  - MAE: the mean absolute error; MASE below 1: better than the seasonal naive method in sample
- Correlogram and Ljung–Box
  - $\hat{\gamma}(h)$: the sample autocovariance at lag h; $\hat{\rho}(h)$: the sample autocorrelation; T: the sample size
  - $Q^*(m)$: the sum of the first m squared autocorrelations, weighted; k: the number of estimated ARMA parameters
  - a Q^* above the $\chi^2(m - k)$ critical value (a small p-value) rejects white noise
- Models and criteria
  - ϕ: the AR(1) coefficient; θ: the MA(1) coefficient; $\ln L$: the maximised log-likelihood; k in AIC and BIC: all estimated parameters
  - ψ_j: the weights of the MA(∞) form of the model ($\psi_0 = 1$); σ: the standard deviation of the shocks; h: the horizon

Prerequisites for Today (3/22): the PACF and model identification

- **PACF** at lag h: the last coefficient ϕ_{hh} of an AR(h) regression

$$y_t = \phi_{h\mathbf{1}} y_{t-\mathbf{1}} + \cdots + \phi_{hh} y_{t-h} + e_t$$

  - the correlation of y_t and y_{t-h} after removing the effect of the lags in between
  - an AR(p) has $\phi_{hh} = 0$ for $h > p$: the PACF cuts off after lag p

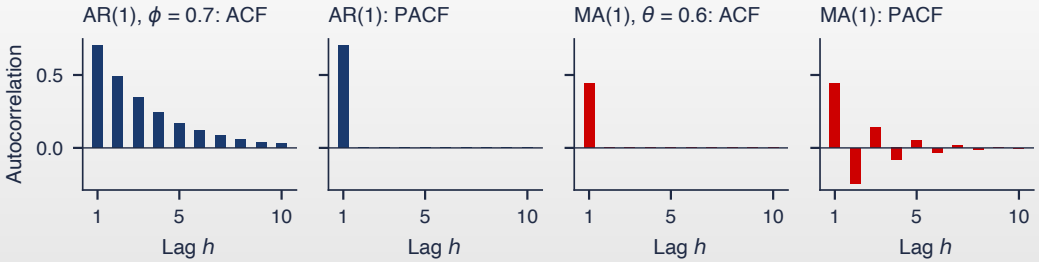
- **Ljung–Box** $Q^*(m)$: are the first m autocorrelations jointly zero?
  - H_0: white noise; p-value $= \Pr(\chi^2(m-k) > Q^*)$, with $m-k$ degrees of freedom on the residuals
  - a p-value below 0.05: the residuals still contain autocorrelation, change the model

- **Information criteria**: AIC and BIC of the formula sheet; for short samples, AICc

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{T-k-1}$$

  - BIC penalises each extra parameter by $\ln T$ instead of 2

Prerequisites for Today (4/22): ACF and PACF in a chart



- AR(1): the ACF ϕ^h decays, the PACF cuts off after lag 1; MA(1): the ACF cuts off after lag 1, the PACF decays with alternating signs
- MA(1) here: $\rho(1) = 0.6/(1 + 0.36) = 0.441$; with data, compare each bar with the band $\pm 1.96/\sqrt{T}$

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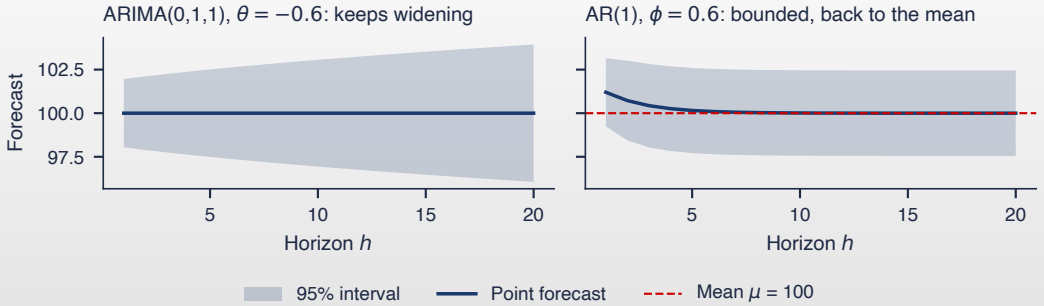
Prerequisites for Today (5/22): forecast intervals and the ψ weights

- Every ARMA or ARIMA model is a sum of past shocks, $y_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$; the **forecast error** at horizon h has variance

$$\mathrm{Var}(y_{T+h} - \hat{y}_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$$

  - only the shocks after the origin T are unknown: $\varepsilon_{T+1}, \ldots, \varepsilon_{T+h}$
- **AR(1)**: $\psi_j = \phi^j$
  - the sum converges to $1/(1-\phi^2)$: the interval stays bounded, the forecast returns to the mean
- **ARIMA(0,1,1)**: $\psi_0 = 1$, $\psi_j = 1 + \theta$ for $j \geq 1$
  - the sum $1 + (h-1)(1+\theta)^2$ grows linearly in h: the interval keeps widening (a unit root)
  - MA part invertible if $|\theta| < 1$: the root $-1/\theta$ of $1 + \theta z$ lies outside the unit circle

Prerequisites for Today (6/22): forecast intervals in a chart



- Left: ARIMA(0,1,1) with $\theta = -0.6$, $\sigma = 1$: a flat forecast, half-width $1.96\sqrt{1 + 0.16(h-1)}$, without limit
- Right: AR(1) with $\phi = 0.6$: the forecast returns to μ, the half-width tends to $1.96/\sqrt{1-0.36} = 2.45$

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Prerequisites for Today (7/22): unit roots, ARIMA, SARIMA, evaluation

Quantity	Formula
ADF	$\Delta y_t = c + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$; H_0: $\gamma = 0$ (unit root)
ADF, 5% critical values	about -2.87 (constant), -3.43 (constant and trend); reject if τ is smaller
KPSS	H_0: stationarity; 5% critical values 0.463 (constant), 0.146 (trend); reject if larger
ARIMA(0,1,1)	$\Delta y_t = \varepsilon_t + \theta \varepsilon_{t-1}$; $\hat{y}_{T+h} = y_T + \theta \varepsilon_T$, $\alpha_{SES} = 1 + \theta$
Its error variance	$\sigma^2[1 + (h-1)(1+\theta)^2]$
AR(1)×SAR(1)$_s$	$z_t = \phi z_{t-1} + \Phi z_{t-s} - \phi\Phi z_{t-s-1} + \varepsilon_t$
Annual rate from z	$z_t = \Delta\Delta_s \ln y_t$: $\Delta_s \ln y_{T+1} = \Delta_s \ln y_T + \hat{z}_{T+1}$
Diebold–Mariano	$d_t = e_{1t}^2 - e_{2t}^2$, DM $= \bar{d}/(s_d/\sqrt{n})$; reject equal accuracy if $\lvert\text{DM}\rvert > t_{0.975}(n-1)$

Prerequisites for Today (8/22): the notation of the previous sheet

- ADF and KPSS
  - $\Delta y_t = y_t - y_{t-\mathbf{1}}$; c: the constant; bt: the deterministic trend; δ_j: the coefficients of the lagged differences
  - $\gamma = 0$: a unit root; τ: the t-statistic of $\hat{\gamma}$, compared with Dickey–Fuller critical values, not Student ones
- ARIMA and SARIMA
  - ε_T: the last shock; α_{SES}: the weight of the equivalent exponential smoothing
  - s: the season length; Φ: the seasonal AR coefficient; $\Delta_s y_t = y_t - y_{t-s}$; z_t: the doubly differenced series
- Diebold–Mariano
  - $e_{\mathbf{1}t}$, $e_{\mathbf{2}t}$: the errors of the two forecasts; d_t: the difference of their squares; $\bar{d}$, s_d: its mean and standard deviation; n: the number of forecasts
  - $t_{\mathbf{0.975}}(n-1)$: the 97.5% quantile of the Student distribution with $n-1$ degrees of freedom

Prerequisites for Today (9/22): unit roots: ADF and KPSS

- **ADF** Dickey and Fuller (1979): subtract y_{t-1} from an AR(1) $y_t = \phi y_{t-1} + \varepsilon_t$

$$\Delta y_t = \gamma\, y_{t-1} + \varepsilon_t, \qquad \gamma = \phi - 1$$

  - $\gamma = 0$ ($\phi = 1$): a random walk, shocks are permanent; $\gamma < 0$: stationary, shocks die out
  - the lagged differences Δy_{t-j} of the sheet absorb the remaining autocorrelation

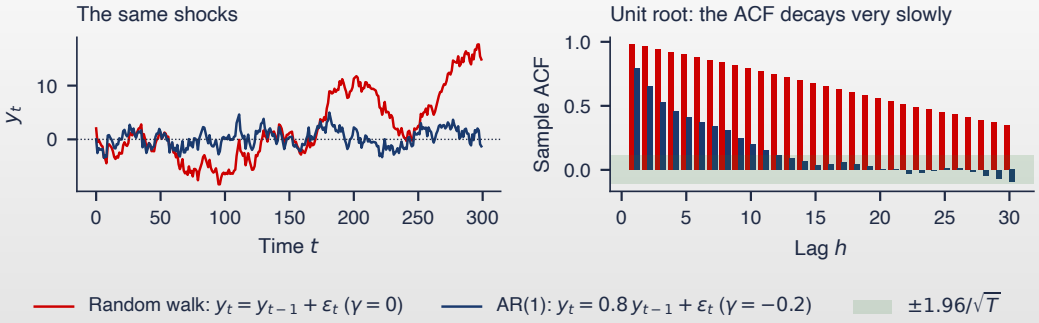
- **KPSS** Kwiatkowski et al. (1992): the opposite null, H_0: stationarity

$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}^2_{LR}} \sum_{t=1}^{T} S_t^2, \qquad S_t = \sum_{s=1}^{t} \hat{e}_s$$

  - $\hat{e}_s$: the residuals of y on a constant (or a trend); S_t: their partial sums; $\hat{\sigma}^2_{LR}$: the long-run variance of $\hat{e}$
  - a unit root makes S_t wander far from 0: KPSS large, reject stationarity

- Together: ADF does not reject and KPSS rejects: $I(1)$; ADF rejects and KPSS does not: $I(0)$; otherwise the evidence is mixed

Prerequisites for Today (10/22): a unit root in a chart



- Left: the same shocks; the random walk wanders without a fixed mean, the AR(1) with $\phi = 0.8$ keeps returning to 0
- Right: the sample ACF of the random walk stays high for 30 lags; $I(1)$: integrated of order 1, stationary after one difference

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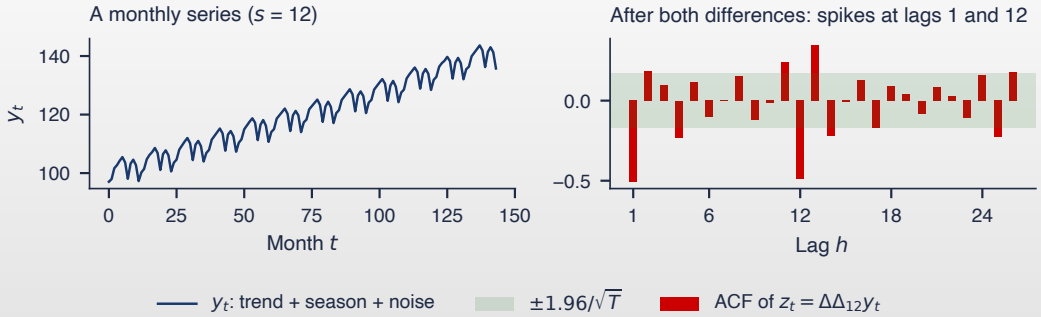
Prerequisites for Today (11/22): seasonal ARIMA and the airline model

- **Seasonal difference** $\Delta_s y_t = (1 - L^s) y_t = y_t - y_{t-s}$; $s = 12$ (monthly), $s = 4$ (quarterly)
  - $z_t = \Delta\Delta_s y_t$ removes the trend and a seasonal pattern that changes slowly
- **SARIMA**$(p, d, q)(P, D, Q)_s$ Box, Jenkins and Reinsel (2008): AR and MA polynomials in L and in L^s
  - the **airline model** $(0, 1, 1)(0, 1, 1)_{\mathbf{12}}$:

$$(1 - L)(1 - L^{\mathbf{12}})\, y_t = (1 + \theta L)(1 + \Theta L^{\mathbf{12}})\, \varepsilon_t$$

  - θ: the MA coefficient at lag 1; Θ: the seasonal MA coefficient at lag 12; z_t has ACF spikes at lags 1, 11, 12 and 13 only
- **Out-of-sample accuracy** on N test months
  - RMSE $= \sqrt{\frac{1}{N}\sum_i e_i^{\mathbf{2}}}$, MAE $= \frac{1}{N}\sum_i |e_i|$; the seasonal naive forecast $\hat{y}_{T+h} = y_{T+h-\mathbf{12}}$ is the benchmark

Prerequisites for Today (12/22): seasonal differencing in a chart



- Left: a simulated monthly series with a trend and a seasonal pattern
- Right: the ACF of $z_t = \Delta\mathbf{\Delta_{12}} y_t$ of an airline model: negative spikes at lags 1 and 12, as $\theta < 0$ and $\Theta < 0$

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Prerequisites for Today (13/22): volatility and several series

Quantity	Formula
GARCH(1,1)	$\sigma^2_{t+1} = \omega + \alpha\varepsilon^2_t + \beta\sigma^2_t$, $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$, $h_{1/2} = \ln 0.5/\ln(\alpha+\beta)$
Annual volatility	$\sqrt{A\,\bar{\sigma}^2}$, $A =$ trading days a year
VaR 1%	$-(\mu + \sigma_{t+1} q_{0.01}(z))$; Student-$t_\nu$: $q = t_\nu^{-1}(0.01)\sqrt{(\nu-2)/\nu}$; Normal: -2.326
VAR(p)	$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t$, OLS equation by equation
Granger F	$[(RSS_R - RSS_U)/p] \,/\, [RSS_U/(T-Kp-1)] \sim F(p, T-Kp-1)$
Long-run effect	$\partial y/\partial x = \sum_j b_{yx,j} \,/\, (1 - \sum_j b_{yy,j})$
Engle–Granger, ECM	ADF on OLS residuals, critical value about -3.34 (two variables); $h_{1/2} = \ln 0.5/\ln(1+\gamma)$
Johansen	trace $-T\sum_{i>r}\ln(1-\hat{\lambda}_i)$; test $r = 0, 1, \ldots$ in turn; stop at the first non-rejection

Prerequisites for Today (14/22): the notation of the previous sheet

- GARCH and VaR
  - $\sigma_t^{\mathbf{2}}$: the conditional variance; ε_t: the shock; ω: the constant; α: the reaction to shocks; β: the persistence
  - $\bar{\sigma}^{\mathbf{2}}$: the long-run variance; $h_{\mathbf{1/2}}$: the half-life of a volatility shock; μ: the mean return
  - $q_{\mathbf{0.01}}(z)$: the 1% quantile of the standardised innovations; ν: the degrees of freedom of the Student-t distribution
- VAR and Granger
  - $\mathbf{y}_t$: the vector of K variables; $\mathbf{c}$: the constants; $\mathbf{A}_i$: the coefficient matrix of lag i; $\mathbf{u}_t$: the shocks; p: the number of lags
  - RSS_R, RSS_U: the residual sums of squares without and with the lags of the tested variable
  - $b_{yx,j}$: the coefficient of x_{t-j} in the equation of y; $b_{yy,j}$: that of y_{t-j}
- Cointegration
  - $\gamma < 0$: the error-correction coefficient, the speed of return to equilibrium
  - $\hat{\lambda}_i$: the estimated eigenvalues of the Johansen procedure; r: the tested cointegration rank

Prerequisites for Today (15/22): GARCH: volatility clustering and the half-life

- **GARCH(1,1)** Bollerslev (1986): today's shock and yesterday's variance give tomorrow's variance
  - $\alpha + \beta < 1$: after a shock the variance returns to $\bar{\sigma}^2$
- **Variance forecast** h days ahead

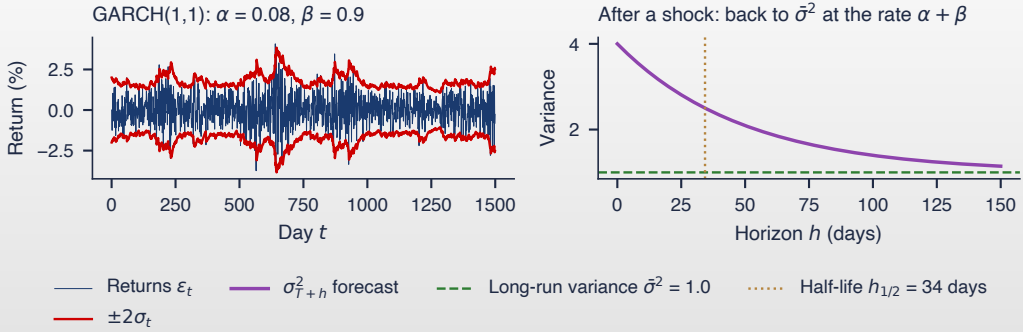
$$\sigma^2_{T+h} - \bar{\sigma}^2 = (\alpha + \beta)^{h-1} (\sigma^2_{T+1} - \bar{\sigma}^2)$$

  - the half-life $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ solves $(\alpha + \beta)^h = 0.5$: the gap to $\bar{\sigma}^2$ halves
- **ARCH-LM test** Engle (1982) with q lags: regress $\hat{\varepsilon}^2_t$ on $1, \hat{\varepsilon}^2_{t-1}, \ldots, \hat{\varepsilon}^2_{t-q}$

$$\text{LM} = T\, R^2 \sim \chi^2(q) \text{ under } H_0\text{: no ARCH effects}$$

  - $\hat{\varepsilon}_t$: the demeaned returns; R^2: of this regression; a large LM: the size of returns is predictable

Prerequisites for Today (16/22): GARCH in a chart



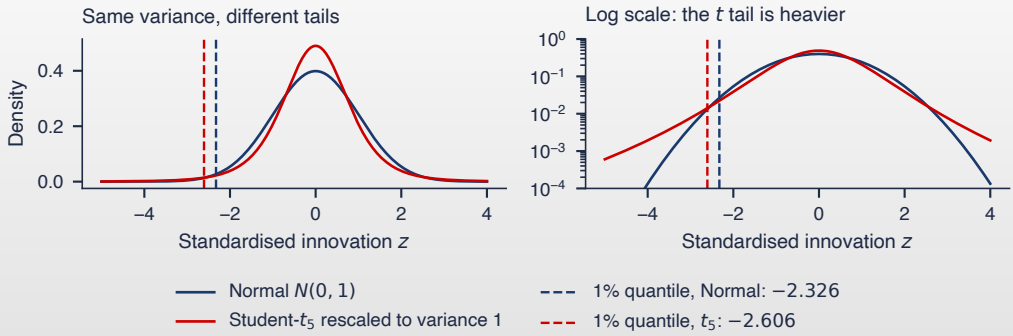
- Left: simulated GARCH(1,1) returns; calm and turbulent periods alternate (volatility clustering), $\pm 2\sigma_t$ follows them
- Right: $\alpha + \beta = 0.98$, so $h_{1/2} = \ln 0.5 / \ln 0.98 = 34$ days: a variance four times $\bar{\sigma}^2$ halves its excess in 34 days

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Prerequisites for Today (17/22): VaR with Normal and Student-t innovations

- Return $r_{t+\mathbf{1}} = \mu + \sigma_{t+\mathbf{1}} z_{t+\mathbf{1}}$, with z of mean 0 and variance 1; **VaR 1%**: the loss exceeded with probability 1%
  - VaR 1% $= -(\mu + \sigma_{t+\mathbf{1}} q_{\mathbf{0.01}}(z))$ (formula sheet) is reported as a positive loss, at the level "1%" (the tail probability)
- **Student-t_ν** innovations: t_ν has variance $\nu/(\nu-2)$, so it is rescaled to variance 1
  - hence the factor $\sqrt{(\nu-2)/\nu}$ in $q_{\mathbf{0.01}}(z)$ of the formula sheet; $t_\nu^{-\mathbf{1}}(0.01)$: the 1% quantile of the Student distribution with ν degrees of freedom; small ν: heavy tails

Prerequisites for Today (18/22): VaR in a chart



- Normal and Student-t_5 with the same variance: the t has more mass in the centre and in the tails
- 1% quantiles: -2.326 (Normal) and $-3.365 \cdot \sqrt{3/5} = -2.606$ (t_5): with $\mu = 0$, $\sigma = 1\%$, VaR 1% is 2.33% or 2.61%

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Prerequisites for Today (19/22): Granger causality and cointegration

- **Granger causality** Granger (1969): x Granger-causes y if the lags of x improve the forecast of y, given the lags of y
  - $H_{\mathbf{0}}$: all $b_{yx,j} = 0$, tested with the F statistic of the sheet; extra predictability, not proof of a causal mechanism
- **Cointegration** Engle and Granger (1987): y_t, x_t are $I(1)$, but a combination is stationary

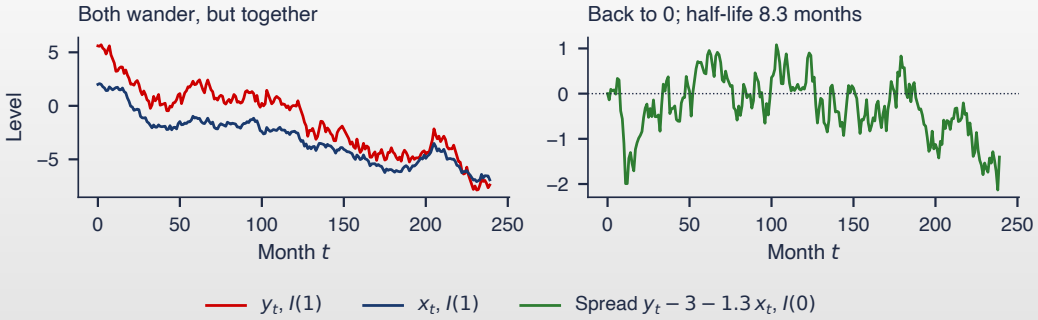
$$u_t = y_t - c - \beta x_t \sim I(0), \qquad \Delta y_t = \gamma\, u_{t-\mathbf{1}} + \delta\, \Delta x_t + \varepsilon_t$$

  - β: the long-run coefficient; the second equation is the **error-correction model** (ECM); δ: the short-run effect
- **Johansen** Johansen (1991): eigenvalues $\hat{\lambda}_{\mathbf{1}} > \hat{\lambda}_{\mathbf{2}} > \dots$ of a VECM; two statistics for rank r

$$\lambda_{trace}(r) = -T \sum_{i>r} \ln(1 - \hat{\lambda}_i), \qquad \lambda_{max}(r) = -T \ln(1 - \hat{\lambda}_{r+\mathbf{1}})$$

  - large values reject "at most r cointegrating relations"; trace: against up to K relations; maximum eigenvalue: against $r + 1$

Prerequisites for Today (20/22): cointegration in a chart



- Left: two simulated $I(1)$ series that wander together; neither has a fixed mean
- Right: the spread $u_t = y_t - 3 - 1.3x_t$ is stationary; with $\gamma = -0.08$, $h_{1/2} = \ln 0.5 / \ln 0.92 = 8.3$ months

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Prerequisites for Today (21/22): extensions

Quantity	Formula
ARFIMA$(0, d, 0)$	$\rho(1) = d/(1-d)$, $H = d + 1/2$; stationary for $d < 1/2$, mean-reverting for $d < 1$
Local level	$F_t = P_t + \sigma_\varepsilon^2$, $K_t = P_t/F_t$, $a_{t\|t} = a_t + K_t(y_t - a_t)$, $P_{t+1} = P_t(1-K_t) + \sigma_\eta^2$
Steady state	$\bar{P}/\sigma_\varepsilon^2 = (q + \sqrt{q^2+4q})/2$, $\bar{K} = \alpha_{SES}$, $q = \sigma_\eta^2/\sigma_\varepsilon^2$
Markov switching	$E(D_i) = 1/(1-p_{ii})$, $\pi_1 = (1-p_{22})/(2-p_{11}-p_{22})$
Machine learning	walk-forward validation only; compare with a simple benchmark
Values	$\chi^2_{0.95}$: 3.84 (1), 5.99 (2), 9.49 (4), 12.59 (6), 18.31 (10); $F_{0.95}(2, 250) = 3.03$

Prerequisites for Today (22/22): the notation of the previous sheet

- Long memory
  - d: the order of fractional integration; H: the Hurst exponent; $\rho(1)$: the first autocorrelation
- Local level model and the Kalman filter
  - a_t, P_t: the predicted level and its variance; $a_{t|t}$: the level updated with y_t
  - σ^2_ε, σ^2_η: the variances of the measurement and level noise; q: their ratio
  - F_t: the variance of the prediction error $y_t - a_t$; K_t: the Kalman gain, the share of the surprise added to the level; $\bar{P}$, $\bar{K}$: their steady-state values
- Markov switching and critical values
  - p_{ii}: the probability of staying in regime i; $E(D_i)$: the expected duration of regime i; π_1: the long-run share of regime 1
  - $\chi^2_{0.95}$: 95% quantiles of the chi-square distribution, degrees of freedom in brackets; $F_{0.95}(2, 250)$: the 95% quantile of the F distribution

A1: an ARIMA(0,1,1) by hand [Solved]

Task

- $\Delta y_t = \varepsilon_t - 0.6\,\varepsilon_{t-1}$, $\varepsilon_t \sim WN(0,1)$; y_0 given, $\varepsilon_0 = 0$; today $y_T = 100$, $\hat{\varepsilon}_T = 1.5$.
- 1. Show that Δy_t is stationary and invertible, and compute $\rho_{\Delta y}(1)$.
- 2. Compute $\mathrm{Var}(y_t)$ for $t = 10$ and $t = 100$ and say whether y_t is stationary.
- 3. Forecast y_{T+1} and y_{T+4}; give the 95% interval for $h = 4$.
- 4. Give the weight of the equivalent exponential smoothing.
- Report: five numbers, an interval and one sentence.

Solution

- 1. An MA(1) is always stationary; root $1/0.6 = 1.667 > 1$: invertible; $\rho(1) = -0.6/1.36 = -0.441$
- 2. $y_t = y_0 + \varepsilon_t + 0.4\sum_{j<t}\varepsilon_j$: $\mathrm{Var} = 1 + (t-1)0.16$: 2.44 and 16.84: grows with t, not stationary
- 3. $\hat{y}_{T+h} = 100 - 0.6 \times 1.5 = 99.10$ for every h; variances 1.00 and 1.48; interval $[96.72, 101.48]$
- 4. $\alpha = 1 + \theta = 0.4$
- Interpretation: the forecast is flat, like that of exponential smoothing, but its interval keeps widening: a unit root

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A2: identifying the Romanian unemployment rate [Proposed]

Solution discussed in class

Task

Series	ADF τ	p	ADF crit. 5%	KPSS (crit. 0.463)
u_t	-1.52	0.524	-2.87	1.037
Δu_t	-8.12	< 0.001	-2.87	0.096

- Unemployment rate u_t (seasonally adjusted, %), January 2005 – August 2026; both tests with a constant. Δu_t ($T = 259$): ACF -0.19, 0.11, 0.09, -0.07; PACF -0.19, 0.08, 0.13, -0.04 (lags 1–4). Model: Seminar 3, A1; Seminar 2, A6.
- 1. State H_0 of each test and decide the order of integration of u_t.
- 2. Compute the band $\pm 1.96/\sqrt{T}$ and Ljung–Box $Q(4)$ from the four ACF values; decide at 5%.
- 3. Propose an ARIMA model for u_t.
- Report: two decisions, two numbers, a model.

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A3: SARIMA output for Romanian GDP [Proposed]

Solution discussed in class

Task

Dependent: z_t	Coef.	SE	z	p
AR(1)	−0.083	0.106	−0.79	0.431
SAR(4)	−0.405	0.085	−4.79	< 0.001
SIGMASQ	10.41			
Ljung–Box $Q(8)$	18.14			0.006

- $z_t = \Delta\Delta_4\, 100 \ln \mathrm{GDP}_t$, real GDP not seasonally adjusted, 2001Q2 – 2026Q2 ($T = 101$). Last values of z: 1.923 (2025Q2), −1.150 (2025Q3), −2.584 (2025Q4), −0.247 (2026Q1), −0.457 (2026Q2); annual growth $\Delta_4\, 100 \ln$ GDP in 2026Q2: −1.96%. Model: Seminar 4, A4.
- 1. Write the model equation and say which coefficients are significant.
- 2. Test whether the residuals are white noise ($\chi^2_{0.95}(6) = 12.59$).
- 3. Forecast z and the annual growth rate for 2026Q3.

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A4: VAR output and Granger causality [Proposed]

Solution discussed in class

Task

	π_t equation	i_t equation
π_{t-1}	1.318 [21.58]	0.031 [0.48]
π_{t-2}	−0.342 [−5.52]	0.008 [0.13]
i_{t-1}	0.011 [0.19]	1.167 [19.60]
i_{t-2}	−0.012 [−0.22]	−0.219 [−3.81]
const.	0.112 [1.62]	0.063 [0.88]
R^2	0.97	0.97

- VAR(2) for Romanian annual inflation π and ROBOR 3M i (%), March 2005 – August 2026, $T = 258$; t in brackets. Without the lags of π, the RSS of the i equation rises from 89.81 to 92.20. Model: Seminar 6, A5.
- 1. Write the i equation.
- 2. Compute the Granger F statistic for "π does not Granger-cause i" and decide at 5% ($F_{0.95} = 3.03$).
- 3. Compute the long-run effect on ROBOR of a permanent 1 pp rise in inflation.

A5: GARCH output for the DAX and VaR 1% [Proposed]

Solution discussed in class

Task

DAX	μ	ω	α	β	ν
estimate	0.081	0.0206	0.097	0.895	7.08
SE	0.012	0.0049	0.011	0.012	0.617

- GARCH(1,1)-t, daily DAX log returns in % since 2000 ($n = 6\,786$); last day (September 2026): $\sigma^2_T = 0.662$, $\hat{\varepsilon}_T = -1.698$; about 254 trading days a year; use $\nu = 7$: $t_7^{-1}(0.01) = -2.998$. Model: Seminar 5, A1 and A4.
- 1. Compute the persistence, the half-life and the long-run annual volatility.
- 2. Compute σ^2_{T+1}.
- 3. Compute the one-day VaR 1% and compare it with the Normal VaR 1%.

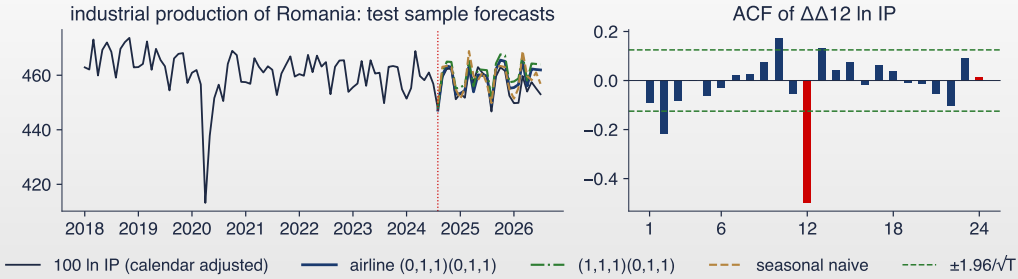
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B1: Box–Jenkins for Romanian industrial production [Solved]

- **Question**: which SARIMA model forecasts Romanian industrial production best, and does it beat the seasonal naive method?
- **Inputs**: industrial production (B–D), calendar adjusted, monthly, January 2005 – July 2026; test sample: the last 24 months
- **Tasks**
  1. Run ADF and KPSS on $100 \ln y_t$, its first difference and $z_t = \Delta\Delta_{\mathbf{12}} 100 \ln y_t$.
  2. Plot the ACF of z_t and propose a model.
  3. Fit the airline model and SARIMA$(1,1,1)(0,1,1)_{\mathbf{12}}$ on the training sample; compare AICc, BIC and Ljung–Box.
  4. Forecast the 24 test months with both models and with the seasonal naive method; compare RMSE and MAE.
  5. Interpretation: which model would you use to forecast, and why?
- **Report**: a table of tests, a table of two models, three RMSE and one sentence
- Notebook: section B1

B1: solution (1/2) [Solved]



Model	AICc	BIC	LB(12) p	LB(24) p	test RMSE
airline $(0,1,1)(0,1,1)_{12}$	1304.2	1314.3	0.065	0.432	3.75
$(1,1,1)(0,1,1)_{12}$	1291.1	1304.5	0.789	0.972	5.29
seasonal naive					3.85

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B1: solution (2/2) [Solved]

- Tests: ADF $\tau = -1.31$ ($p = 0.885$) and KPSS $0.523 > 0.146$ for the level; z_t: ADF -6.47, KPSS 0.055: $d = D = 1$; ACF of z_t: -0.09, -0.22 at lags 1, 2, -0.50 at 12
- Airline: $\hat{\theta} = -0.205$, $\hat{\Theta} = -0.934$; the $(1, 1, 1)$ model: $\hat{\phi} = 0.574$, $\hat{\theta} = -0.828$, almost cancelling factors
- Interpretation: the $(1, 1, 1)$ model wins in sample (BIC, clean residuals) but loses out of sample; the airline model is slightly better than the seasonal naive method: use the airline model, and keep the seasonal naive method as the benchmark

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B2: volatility of the BET since 2015 [Proposed]

- **Question**: is the Bucharest market calmer or more turbulent today than usual, and what does it mean for tomorrow's VaR 1%? Model: Seminar 5, B1.
- **Inputs**: BET daily log returns in %, January 2015 – September 2026 ($n = 2\,925$)
- **Tasks**
  1. Run the ARCH-LM test with five lags on the demeaned returns.
  2. Fit a GARCH(1,1) with Student-t innovations; report the persistence and the half-life.
  3. Compare the long-run, the sample and today's annualised volatility.
  4. Compute tomorrow's VaR 1% with the t and with the Normal quantile.
  5. Interpretation: is the BET calmer or more turbulent than usual today?
- **Report**: one test, five parameters, three volatilities, two VaR values and one sentence
- Notebook: section B2

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B3: the Romanian and German 10-year yields [Proposed]

- **Question**: is the Romanian 10-year yield tied to the German one in the long run? Model: Seminar 7, B1 and A3.
- **Inputs**: monthly means of daily 10-year yields (%), January 2008 – September 2026 ($T = 224$)
- **Tasks**
  1. Run ADF on both yields and on their differences.
  2. Run the Engle–Granger test (Romania on Germany) and report $\hat{\beta}$.
  3. Run the Johansen trace and maximum-eigenvalue tests.
  4. Estimate the error-correction equation of each yield and compute the half-life.
  5. Interpretation: is the Romanian yield tied to the German one?
- **Report**: four ADF results, two cointegration tests, two adjustment coefficients and one sentence
- Notebook: section B3

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C1: a forecast competition for Romanian inflation [Proposed]

- **Question**: which method would have forecast Romanian monthly inflation best over the last ten years, one and twelve months ahead?
- **Inputs**: the Romanian HICP (Eurostat), monthly, 2005–2026; Lecture 15 runs one SARIMA on one test sample
- **Tasks**
  1. List five competitors: seasonal naive, SES, SARIMA, ARFIMA, a ridge model with lags and month dummies.
  2. Describe a rolling-origin design: first origin, step, horizons, refits.
  3. Say how you would treat the tax changes of 2010, 2015 and 2025.
  4. Choose the error measures and the test for the comparison.
- **Report**: a one-page plan that a team could turn into its project
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to check the answers to Part A. The answer:
  - `(a) For the unemployment rate, ADF gives p = 0.52, so the unit root is rejected at 5%.`
  - `(b) KPSS = 1.04 > 0.463 rejects, so the level of unemployment is stationary.`
  - `(c) For the DAX, alpha + beta = 0.992, so a shock halves in 1/(1 - 0.992) = 128 days.`
  - `(d) The VaR 99% of the DAX for tomorrow is -2.31%.`
  - `(e) Inflation Granger-causes ROBOR (p = 0.036), which proves that inflation causes the central bank to raise rates.`
  - `(f) Ljung-Box Q(12) on the residuals of an AR(1) x SAR(1) model has 12 degrees of freedom.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and the correct number (notebook, section C2).
  2. Report: six verdicts with one line of justification each.

Key takeaways

- Every exam problem has three parts: the formula with the numbers, the result with its unit and sign, the interpretation
- Tests: state H_0 first; ADF and KPSS have opposite nulls; Ljung–Box on residuals loses k degrees of freedom
- Forecasts by hand: write the model in levels of z_t, substitute the last values, then undo the differences
- A better in-sample model is not a better forecaster (B1); judge it on a test sample against a benchmark
- An AI answer is a draft: check the null hypotheses, the half-life, the VaR level and the degrees of freedom

After the seminar

- Lecture 15 gathers the course: the course map, two recap slides per chapter, the toolbox, Box–Jenkins on Romanian inflation, eight solved exam-type problems, the project
- Try the [Proposed] tasks on paper first, then check them in the notebook
- C1 can grow into a team project
- More practice: the practice set of 25 exam problems with numerical answers
- Reading: Huang and Petukhina (2022); Hyndman and Athanasopoulos (2021); Box, Jenkins and Reinsel (2008)

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